

Technical Analysis of VLSM & Supernetting

A structured, exam-oriented synthesis with implementation logic and visual grounding

1. Conceptual Big Picture

CIDR-based IP management relies on **two complementary operations**:

- **VLSM (Variable Length Subnet Masking)** → *divide downward*
One large block → multiple **unequal** subnets
- **Supernetting (Route Aggregation)** → *merge upward*
Multiple equal-sized networks → one **larger** CIDR prefix

These techniques exist for **two different optimization goals**:

Technique	Optimizes	Where Used
VLSM	Address utilization	Inside organizations
Supernetting	Routing table size	Between routers / ISPs

2. VLSM: Technical Deep Dive

Core Principle

VLSM allows **different prefix lengths inside the same parent block**.

Mathematically:

- Host bits = $32 - \text{prefix}$
- Block size = $2^{(\text{host bits})}$
- Usable hosts = $2^{(\text{host bits})} - 2$

Case Study Recap

Given block: 14.24.74.0/24

Total IPs: 256

Requirements:

- 120 IPs
- 60 IPs
- 10 IPs

Optimal Allocation Strategy (Largest First)

Subnet	Needed IPs	Host Bits	Prefix	Block Size	Range
A	120	7	/25	128	.0 – .127
B	60	6	/26	64	.128 – .191
C	10	4	/28	16	.192 – .207

Why this works

- Each block size is a **power of two**
- Each block starts at an address **divisible by its size**
- No overlaps, no gaps
- Minimal waste

Key VLSM Rule:

Always allocate the largest subnet first.

Failing this almost always leads to fragmentation or invalid alignment.

Why Fixed-Length Subnetting Fails Here

If all subnets were forced to /25 or /26:

- The 10-host subnet would waste **50+ IPs**
- Future growth would be constrained

VLSM prevents this by **customizing prefix length per requirement**.

3. Supernetting (Route Aggregation): Technical Analysis

Purpose

Supernetting reduces **routing table entries**, not address waste.

Routers prefer:

- **Fewer prefixes**
- **Shorter lookup times**
- **Smaller memory footprint**

4. The Four Mandatory Rules of Supernetting

All **four must be satisfied** — no exceptions.

1 Contiguous Network IDs

Networks must be numerically adjacent.

128.56.24.0
128.56.25.0
128.56.26.0
128.56.27.0

2 Identical Network Sizes

All networks must have the **same prefix**.

- ✓ All /24
- ✗ /24 mixed with /23 (unless staged aggregation)

3 Power-of-Two Quantity

Number of networks must be 2^n .

- ✓ 2, 4, 8, 16
- ✗ 3, 5, 6

4 Divisibility Rule (Most Common Failure Point)

The **first network ID** must be divisible by the **total supernet size**.

This ensures:

- All host bits = 0
- The supernet can be represented by a **single CIDR prefix**

5. Supernet Example Explained

Aggregating Four /24 Networks

128.56.24.0/24

128.56.25.0/24

128.56.26.0/24

128.56.27.0/24

Step 1: Total Size

- Each /24 = 256 IPs
- $4 \times 256 = 1024 = 2^{10}$

Step 2: New Prefix

- Host bits = 10
- Prefix = $32 - 10 = /22$

Step 3: Mask

/22 $\rightarrow$ 255.255.252.0

Step 4: Divisibility Check

- Third octet block size = $256 - 252 = 4$
- Starting octet = 24
- $24 \div 4 = 6 \rightarrow$ integer ✓

Valid Supernet: 128.56.24.0/22

6. The “AD Rule” (High-Speed Exam Shortcut)

When aggregating:

- **Supernet ID** = first network's ID
- **New prefix** = original prefix – $\log_2(\text{number of networks})$

Example:

- 4 networks $\rightarrow \log_2(4) = 2$
- /24 – 2 = /22

This shortcut works **only if divisibility is satisfied**.

7. Subnetting vs Supernetting (Side-by-Side)

Feature	Subnetting	Supernetting
Direction	One $\rightarrow$ Many	Many $\rightarrow$ One
Bit Borrowing	From Host ID	From Network ID
Prefix Change	Increases	Decreases
Goal	Reduce IP waste	Reduce routing entries
Typical Scope	Internal LAN design	ISP / backbone routing

8. Advanced & Exam-Relevant Scenarios

Non-Uniform Aggregation

When prefixes differ:

- Aggregation must be **staged**
- Only blocks of equal size can merge
- If divisibility fails $\rightarrow$ **multiple routes advertised**

This is common in:

- Competitive exams (e.g., GATE)
- Real-world ISP routing policies

9. Final Technical Insights

VLSM

- Solves **address exhaustion**
- Requires careful block alignment
- Largest-first allocation is mandatory

Supernetting

- Solves **routing scalability**
- Divisibility rule is decisive
- CIDR prefix must represent **all networks exactly**

Mental Model

- **Subnetting**: zoom in
- **Supernetting**: zoom out